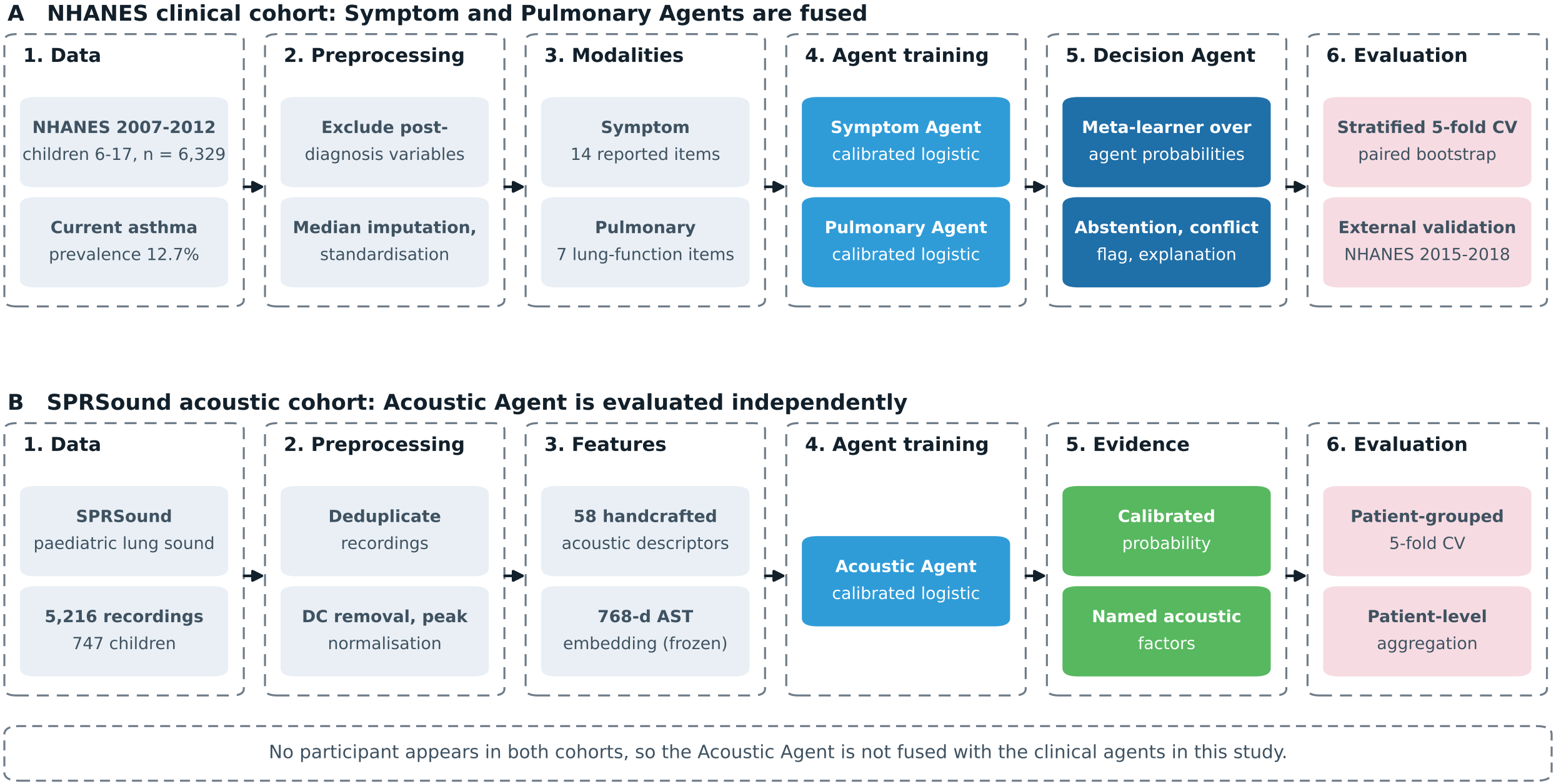
A NHANES clinical cohort: Symptom and Pulmonary Agents are fused
1. Data
NHANES 2007-2012
children 6-17, n = 6,329
Current asthma
prevalence 12.7%
2. Preprocessing
Exclude post-
diagnosis variables
Median imputation,
standardisation
3. Modalities
Symptom
14 reported items
Pulmonary
7 lung-function items
4. Agent training
Symptom Agent
calibrated logistic
Pulmonary Agent
calibrated logistic
5. Decision Agent
Meta-learner over
agent probabilities
Abstention, conflict
flag, explanation
6. Evaluation
Stratified 5-fold CV
paired bootstrap
External validation
NHANES 2015-2018
B SPRSound acoustic cohort: Acoustic Agent is evaluated independently
1. Data
SPRSound
paediatric lung sound
5,216 recordings
747 children
2. Preprocessing
Deduplicate
recordings
DC removal, peak
normalisation
3. Features
58 handcrafted
acoustic descriptors
768-d AST
embedding (frozen)
4. Agent training
Acoustic Agent
calibrated logistic
5. Evidence
Calibrated
probability
Named acoustic
factors
6. Evaluation
Patient-grouped
5-fold CV
Patient-level
aggregation
No participant appears in both cohorts, so the Acoustic Agent is not fused with the clinical agents in this study.